

Section 5.1 If $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies that

$$|f(x) - f(y)| \leq (x-y)^2 \text{ for all } x, y \in \mathbb{R}$$

Then f is constant.

Proof. Since $\left| \frac{f(x) - f(y)}{x-y} \right| \leq |x-y|$ for all $x, y \in \mathbb{R}$,

Given $x \in \mathbb{R}$, $f'(x)$ exists with $f'(x) = 0$.

Now, by the Mean Value Theorem, for any $a < b$, there exists $c \in (a, b)$ s.t.

$$f(a) - f(b) = (a-b) \cdot f'(c) = 0$$

Thus f is constant.

Section 5.2. Let $f: (a, b) \rightarrow \mathbb{R}$ be given. If $f'(x) > 0$ for all $x \in (a, b)$, then f is injective.

Moreover, the inverse $f^{-1}: f[(a, b)] \rightarrow (a, b)$ exists and differentiable with

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} \quad (a < x < b)$$

Proof. Let $a < x < y < b$. Then, by the Mean-Value Theorem, there exists $c \in (x, y)$ s.t.

$$f(y) - f(x) = f'(c) \cdot (y-x) > 0$$

Therefore, f is strictly increasing, thus injective. Now, the restriction

$$f: (a, b) \rightarrow f[(a, b)]$$

is bijection, thus there exists a inverse

$$f^{-1}: f[(a, b)] \rightarrow (a, b): f(x) \mapsto x$$

Now, $(f^{-1} \circ f)(x) = x$ implies that

Section 5.3. Let $g: \mathbb{R} \rightarrow \mathbb{R}$ be given with $|g'| \leq M$ for some $M > 0$.

Fix $\varepsilon > 0$, and define $f(x) = x + \varepsilon \cdot g(x)$. prove that f is injective if ε is small enough.

Proof. If $\varepsilon < \frac{1}{M}$, then $f'(x) = 1 + \varepsilon \cdot g'(x) > 0$. Therefore f is injective.

Section 5.4. If $C_0, \dots, C_n \in \mathbb{R}$ satisfies

$$C_0 + \frac{C_1}{2} + \frac{C_2}{3} + \dots + \frac{C_{n-1}}{n} + \frac{C_n}{n+1} = 0$$

, then the polynomial

$$C_0 + C_1 x + C_2 x^2 + \dots + C_{n-1} x^{n-1} + C_n x^n = p(x)$$

has a real root in $(0, 1)$.

Proof. Define $g(x) = C_0 x + \frac{C_1}{2} x^2 + \dots + \frac{C_{n-1}}{n} x^n + \frac{C_n}{n+1} x^{n+1}$

Then, $g(0) = 0$ clearly, and $g(1) = 0$ by the assumption.

By the Mean-Value Theorem, there exists $\alpha \in (0, 1)$ s.t.

$$0 = g(1) - g(0) = g'(\alpha) \cdot (1 - 0) = g'(\alpha)$$

Directly, $p(x) = g'(x)$. therefore $g'(\alpha) = p(\alpha) = 0$.

Section 5.5 Let $f: (0, \infty) \rightarrow \mathbb{R}$ be differentiable.

If $f(x) \rightarrow 0$ as $x \rightarrow \infty$, then $f(x+1) - f(x) \rightarrow 0$ as $x \rightarrow \infty$.

Proof. By the MVT, for any $x > 0$, there exists $c_x \in (x, x+1)$ s.t.

$$f(x+1) - f(x) = f'(c_x) \cdot (1 - 0) = f'(c_x).$$

Let $\varepsilon > 0$ be given. Then, there exists $\delta > 0$ s.t. $x \geq \delta \Rightarrow |f(x)| < \varepsilon$.

For this $\delta > 0$, for any $x \geq \delta$, there is $c_x \in (x, x+1)$ s.t.

$$|f(x+1) - f(x)| = |f'(c_x)| < \varepsilon$$

Thus proved.

Section 5.6. Suppose that $f: (0, \infty) \rightarrow \mathbb{R}$ satisfies that:

- 1) f is continuous for $x \geq 0$
- 2) f' exists for $x > 0$
- 3) f' increases monotonically.
- 4) $f(0) = 0$.

Then, $g: (0, \infty) \rightarrow \mathbb{R}; x \mapsto \frac{f(x)}{x}$ increases monotonically.

Proof. Since $g(x) = \frac{x f(x) - f(x^2)}{x^2}$, enough to show $f(x^2) \geq \frac{f(x)}{x} = g(x)$.

observe that $g(x) = \frac{f(x)}{x} = \frac{f(x) - f(0)}{x - 0} = f'(c_x)$ for some $0 < c_x < x$, by the MVT.

Now, since f' increases monotonically, $f'(x) \geq g(x) = f'(c_x)$.

Section 5.7. Let $f, g: \mathbb{R} \rightarrow \mathbb{R}$ be given.

Suppose that f', g' exist and $g'(x) \neq 0$ for all $x \in \mathbb{R}$, and $f(x) = g(x) = 0$.

Prove $\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \frac{f'(x)}{g'(x)}$

Proof. Since f', g' exist, and $f(0) = g(0) = 0$,

$$\lim_{t \rightarrow x} \frac{f(t)}{g(t)} = \lim_{t \rightarrow x} \frac{\frac{f(t) - f(x)}{t - x}}{\frac{g(t) - g(x)}{t - x}} = \frac{f'(x)}{g'(x)}$$

Section 5.8. $\sum_{n=1}^{\infty} \sin\left(\frac{1}{n}\right)$ converges?

$\frac{\sin \frac{1}{n}}{\frac{1}{n}} \rightarrow 1$ but $\sum \frac{1}{n}$ diverges, thus

Section 5.8. Let $f: [a, b] \rightarrow \mathbb{R}$ be given. Suppose f' exists and is continuous on $[a, b]$.

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\forall x, t \in [a, b], |x - t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(x) \right| < \varepsilon$

Proof. Since f' is continuous on compact, it is continuous uniformly.

Given $\varepsilon > 0$, there exists $\delta > 0$ s.t. $\forall x, t \in [a, b],$

$$|x - t| < \delta \Rightarrow |f'(x) - f'(t)| < \varepsilon$$

Meanwhile, by the Mean-Value Thm,

$$f'(s) = \frac{f(x) - f(t)}{x - t} \quad \text{for some } s \text{ between } x \text{ and } t.$$

$$\text{Now, } |x - t| < \delta \Rightarrow \left| \frac{f(x) - f(t)}{x - t} - f'(t) \right| = |f'(s) - f'(t)| < \varepsilon$$

because $|s - t| \leq |x - t|$.

[Section 5.9.

$$f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} x+3 & x \geq 0 \\ 3 & x < 0 \end{cases}$$

[Section 5.10. Let $f, g: (0,1) \rightarrow \mathbb{C}$ be differentiable such that

$$\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} g(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow 0} f'(x), \lim_{x \rightarrow 0} g'(x) (\neq 0) \quad \text{exist. Then,}$$

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \lim_{x \rightarrow 0} \frac{f'(x)}{g'(x)}$$

Proof. Observe that $\frac{f(x)}{g(x)} = \left[\frac{f(x)}{x} - \lim_{t \rightarrow 0} \frac{f(t)}{t} \right] \cdot \frac{x}{g(x)} + \lim_{t \rightarrow 0} \frac{f(t)}{t} \cdot \frac{x}{g(x)}$

Meanwhile, by the L'Hopital Law,

$$\lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} \left[\frac{\operatorname{Re} f(x)}{x} + i \cdot \frac{\operatorname{Im} f(x)}{x} \right]$$

$$= \lim_{x \rightarrow 0} \left[\operatorname{Re} f'(x) + i \cdot \operatorname{Im} f'(x) \right] = \lim_{x \rightarrow 0} f'(x)$$

Similarly $\lim_{x \rightarrow 0} \frac{x}{g(x)} = \left(\lim_{x \rightarrow 0} g'(x) \right)^{-1}$. Now,

$$\lim_{x \rightarrow 0} \frac{f(x)}{g(x)} = \left(\lim_{x \rightarrow 0} f'(x) \right) \cdot \left(\lim_{x \rightarrow 0} g'(x) \right)^{-1}$$

Section 5.11.

Suppose that f is defined on a neighborhood of $x \in \mathbb{R}$, and $f''(x)$ exists. Prove that

$$\lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} = f''(x)$$

Proof. By the L'Hopital Law,

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{f(x+h) + f(x-h) - 2f(x)}{h^2} &\stackrel{①}{=} \lim_{h \rightarrow 0} \frac{f'(x+h) - f'(x-h)}{2h} \stackrel{②}{=} \frac{1}{2} \cdot \lim_{h \rightarrow 0} \left[\frac{f'(x+h) - f'(x)}{h} + \frac{f'(x-h) - f'(x)}{-h} \right] \\ &= \frac{1}{2} \cdot 2f''(x). \end{aligned}$$

Define $f: \mathbb{R} \rightarrow \mathbb{R}; x \mapsto \begin{cases} 0 & \text{if } x < 0 \\ x^2 & \text{if } x \geq 0 \end{cases}$

Then, $f''(0)$ does not exist, but

$$\lim_{h \rightarrow 0} \frac{h^2 + 0 - 0}{h^2} = 1$$

exists.

$$\frac{1}{2} \cdot \lim_{h \rightarrow 0} (f''(x+h) + f''(x-h)) = f''(x)$$

For all Try

By the Mean-Value Theorem, for any $h > 0$,

$f(x+h) - f(x) = f'(t_1)h$ and $f(x-h) - f(x) = f'(t_2)h$ for some
for some $t_1 \in (x, x+h)$, $t_2 \in (x-h, x)$

$$\frac{f'(t_1) + f'(t_2)}{h} = 2f''(x)$$

Section 5.13. Let $a \in \mathbb{R}$, $c > 0$. Define

$$f: [-1, 1] \rightarrow \mathbb{R}: x \mapsto \begin{cases} 0 & \text{if } x=0 \\ x^a \cdot \sin\left(\frac{1}{|x|^c}\right) & \text{if } x \neq 0 \end{cases}$$

Equivalence Condition Table	f	f'	f''
Continuous	$a > 0$	$a > 1+c$	$a > 2+2c$
bounded	$a \geq 0$	$a \geq 1+c$	$a \geq 2+2c$
Differentiable at $x=0$.	$a > 1$	$a > 2+c$	—
Bounded Variation.	$a > c$	—	—

(-1)

Proof. (1,1). If $a > 0$, then $|f(x)| \leq |x|^a \rightarrow 0$ as $x \rightarrow 0$, therefore continuous.

If $a \leq 0$, put $x_n = \frac{c}{\sqrt[2n+\frac{1}{2}}{\pi}}$. Clearly $x_n \rightarrow 0$ as $n \rightarrow \infty$, but for any $n \in \mathbb{N}$,

$$f(x_n) = \left(\frac{1}{(2n+\frac{1}{2})\pi}\right)^{\frac{a}{c}} \cdot \underbrace{\sin\left(2n\pi + \frac{\pi}{2}\right)}_{=1} = \left(\frac{1}{(2n+\frac{1}{2})\pi}\right)^{\frac{a}{c}}$$

is 1 when $a=0$, and diverges when $a < 0$. Except $x \neq 0$, clearly continuous.

Proof. (2,1) Trivial.

Proof. (3,1) $f(0)$ exists if and only if $\lim_{x \rightarrow 0} \frac{f(x) - f(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} x^{a-1} \cdot \sin\left(\frac{1}{|x|^c}\right)$ exists.

Trivially, the limit exists iff $a-1 > 0$, or $a > 1$.

Proof. (4,1). Suppose that $a \leq c$. Take a partition P_n on $[0,1]$ as

$$P_n \stackrel{\text{def}}{=} \left\{ 0, \frac{c}{\sqrt[2]{n\pi}}, \frac{c}{\sqrt[2]{(n-1)\pi}}, \dots, \frac{c}{\sqrt[2]{2\pi}}, 1 \right\}$$

Then, $V(f, P_{2n+1}) \geq \sum_{k=2}^{n+1} \left(\frac{2}{(2k+1)\pi}\right)^{\frac{a}{c}} \rightarrow \infty$ as $n \rightarrow \infty$ ($a \leq c$, or $\frac{a}{c} \leq 1$)

therefore f is not bounded variation on $[0,1]$ so is $[-1,1]$.

Suppose that $a > c$. (will be continue)

Proof (1,2) f' continuous at $x=0$ if and only if $\lim_{x \rightarrow 0} f'(x) = f'(0)$
 (Trivially, the existence of derivative is assumed.)

If $\alpha > 1+c$, then $f(x) \rightarrow 0$ as $x \rightarrow 0$, and the limit $\lim_{x \rightarrow 0} f'(x)$ exists, therefore

$$f'(0) = \lim_{x \rightarrow 0} \frac{f(x)}{x} = \lim_{x \rightarrow 0} f'(x)$$

by the L'Hôpital Law.

If $\alpha \leq 1+c$, then $f'(x) = \alpha x^{\alpha-1} \cdot \sin\left(\frac{1}{x^c}\right) - C \cdot x^{\alpha-(1+c)} \cdot \cos\left(\frac{1}{x^c}\right)$ ($x > 0$)
 does not converge when $x \rightarrow 0$. (The first term converges to 0 by $\alpha > 0$,
 but the second term does not.)

Proof (2,2) (In this case, the existence of derivative at $x=0$ is not needed.)

Observe that: for $x > 0$,

$$\begin{aligned} f'(x) &= \alpha x^{\alpha-1} \sin\left(\frac{1}{x^c}\right) + x^{\alpha} \cdot (-c) \cdot x^{-c-1} \cdot \cos\left(\frac{1}{x^c}\right) \\ &= \alpha x^{\alpha-1} \sin\left(\frac{1}{x^c}\right) - C \cdot x^{\alpha-(1+c)} \cdot \cos\left(\frac{1}{x^c}\right) \end{aligned}$$

is bounded iff $\alpha - (1+c) \geq 0$, or $\alpha \geq 1+c$.

for $x < 0$,

$$\begin{aligned} f'(x) &= \alpha x^{\alpha-1} \cdot \sin\left(\frac{1}{(-x)^c}\right) + x^{\alpha} \cdot (-x)^{-c-1} \cdot C \cdot \cos\left(\frac{1}{(-x)^c}\right) \\ &= \alpha x^{\alpha-1} \cdot \sin\left(\frac{1}{(-x)^c}\right) + C \cdot x^{\alpha} \cdot (-x)^{-c-1} \cdot \cos\left(\frac{1}{(-x)^c}\right) \\ &= \alpha x^{\alpha-1} \cdot \sin\left(\frac{1}{(-x)^c}\right) + C \cdot (-x)^{\alpha-c-1} \cdot \cos\left(\frac{1}{(-x)^c}\right) \end{aligned}$$

(The values exists only $\alpha = \frac{2n}{m}$ where $n \in \mathbb{N}$, $m \in \mathbb{Z} \setminus \{0\}$, $\gcd(2n, m) = 1$).

is bounded iff $\alpha - c - 1 \geq 0$, or $\alpha \geq 1+c$.

Proof (3,2) $f''(0)$ exists if and only if $\lim_{x \rightarrow 0} \frac{f'(x) - f'(0)}{x}$ exists.

Assume f' is continuous at $x=0$. That is, $\alpha > 1+c$. Then, clearly, $f'(0) = \lim_{x \rightarrow 0} f'(x) = 0$.

Now,
$$\frac{f'(x) - f'(0)}{x} = \alpha x^{\alpha-2} \sin\left(\frac{1}{x^c}\right) - C \cdot x^{\alpha-c-2} \cdot \cos\left(\frac{1}{x^c}\right) \quad (x > 0)$$

has a limit if and only if $\alpha - c - 2 \geq 0$, or $\alpha \geq 2+c$.

(The Case of $x < 0$ is similar)

Proof (1,3) f'' is continuous at $x=0$ iff $\lim_{x \rightarrow 0} f''(x) = f''(0)$.

Observe that for $x > 0$,

$$f''(x) = a(a-1) \cdot x^{a-2} \cdot \sin\left(\frac{1}{x^c}\right) - ac \cdot x^{a-c-2} \cdot \cos\left(\frac{1}{x^c}\right) \\ - c \cdot (a - (1+c)) \cdot x^{a-(2+c)} \cdot \cos\left(\frac{1}{x^c}\right) - c^2 \cdot x^{a-(2+2c)} \cdot \sin\left(\frac{1}{x^c}\right)$$

If $a > 2+2c$, or $a - (2+2c) > 0$, then $f''(x) \rightarrow 0$ as $x \rightarrow 0$, therefore

$$f''(0) = \lim_{x \rightarrow 0} \frac{f'(x)}{x} = \lim_{x \rightarrow 0} f''(x)$$

by the L'Hopital Law (clearly in this case, $f'(x) \rightarrow 0$ as $x \rightarrow 0$)

If $a \leq 2+2c$, then the limit $\lim_{x \rightarrow 0} f''(x)$ does not exist.

Proof (2,3). Similarly $f'(x)$ is bounded nearby $x=0$ if and only if $a - (2+2c) \geq 0$.

Section 5.15. Let $f: (a, \infty) \rightarrow \mathbb{R}$ be twice-differentiable.

Put $M_k = \sup_{x \in (a, \infty)} |f^{(k)}(x)|$, $k=0, 1, 2$. Then $M_1^2 \leq 4M_0M_2$.

Proof. Given $x > a$ and $h > 0$,

$$f(x+2h) = f(x) + f'(x) \cdot 2h + \underbrace{f''(t) \cdot \frac{2h^2}{2!}}_{= \frac{2h^2}{2!}} \text{ for some } t \in (x, x+2h)$$

$$\text{Therefore } |f'(x)| \leq \left| \frac{f(x+2h) - f(x)}{2h} - h \cdot f''(t) \right| \leq \frac{M_0}{h} + M_2 h$$

Take $h = \sqrt{\frac{M_0}{M_2}}$. Then,

$$\forall x > a, |f'(x)| \leq \sqrt{M_0 M_2} + \sqrt{M_0 M_2} = 2\sqrt{M_0 M_2}$$

which implies that $M_1 \leq 2\sqrt{M_0 M_2}$.

Section 5.16. Let $f: (0, \infty) \rightarrow \mathbb{R}$ be twice-differentiable,

f'' is bounded,

$f(x) \rightarrow 0$ as $x \rightarrow \infty$.

Then, $f'(x) \rightarrow 0$ as $x \rightarrow \infty$.

Proof. Since for any $a > 0$,

$$0 \leq \sup_{x \in (a, \infty)} |f'(x)| \leq 2 \sqrt{\sup_{x \in (a, \infty)} |f(x)| \cdot M} \quad (\text{where } M \text{ is bound for } |f''|)$$

Letting $a \rightarrow \infty$ we obtain the result.

Section 5.17. Let $f: [-1, 1] \rightarrow \mathbb{R}$ be three-times differentiable s.t.
 $f(-1)=0$, $f(0)=0$, $f(1)=1$, $f'(0)=0$.

Then, $f^{(3)}(x) \geq 3$ for some $x \in (-1, 1)$.

Proof. By the Taylor's Theorem,

$$f(1) = f(0) + f'(0) + \frac{f''(0)}{2!} + \frac{f^{(3)}(s)}{3!} \quad \text{for some } s \in (0, 1)$$

and $f(-1) = f(0) - f'(0) + \frac{f''(0)}{2!} - \frac{f^{(3)}(t)}{3!} \quad \text{for some } t \in (-1, 0)$

Combining above, then

$$f(1) - f(-1) = 2f'(0) + \frac{1}{3!} (f^{(3)}(s) + f^{(3)}(t))$$

Thus $f^{(3)}(s) + f^{(3)}(t) = 6$, therefore either one bigger than 3.

Q. Why $f'(0)=0$ is needed?

Section 5.18.

Let $f: [a, b] \rightarrow \mathbb{R}$, $n \in \mathbb{N}$, $\alpha, \beta \in [a, b]$. Suppose $f^{(n-1)}$ exists on $[a, b]$. Define

$$Q(t) = \frac{f(t) - f(\beta)}{t - \beta} \quad (t \neq \beta).$$

Then
$$f(\beta) = \sum_{k=0}^{n-1} \frac{f^{(k)}(\alpha)}{k!} \cdot (\beta - \alpha)^k + \frac{Q^{(n-1)}(\alpha)}{(n-1)!} \cdot (\beta - \alpha)^n$$

Proof. Derivating $(n-1)$ times to:

$$(t - \beta) \cdot Q(t) = f(t) - f(\beta)$$

Then,

$$Q(t) + (t - \beta) Q'(t) = f'(t)$$

$$Q'(t) + Q''(t) + (t - \beta) Q''(t) = f''(t)$$

;

$$(n-1) \cdot Q^{(n-2)}(t) + (t - \beta) \cdot Q^{(n-1)}(t) = f^{(n-1)}(t)$$

Now,
$$f(\beta) = f(t) + Q(t) \cdot (\beta - t)$$

$$= f(t) + f'(t) \cdot (\beta - t) + Q'(t) (\beta - t)^2$$

$$= f(t) + f'(t) (\beta - t) + \frac{f''(t)}{2!} (\beta - t)^2 + \frac{Q''(t)}{2!} (\beta - t)^3$$

$$= \sum_{k=0}^{n-1} \frac{f^{(k)}(t)}{k!} \cdot (\beta - t)^k + \frac{Q^{(n-1)}(t)}{(n-1)!} \cdot (\beta - t)^n.$$

Section 5. 1a. Let $f: (-1, 1) \rightarrow \mathbb{R}$ be a function such that $f'(0)$ exists.

Suppose $-1 < \alpha_n < \beta_n < 1$ and $\lim_{n \rightarrow \infty} \alpha_n = \lim_{n \rightarrow \infty} \beta_n = 0$. Define the difference quotient:

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n}$$

1) If $\alpha_n < 0 < \beta_n$, then $\lim_{n \rightarrow \infty} D_n = f'(0)$

2) If $0 < \alpha_n < \beta_n$ and $\left\{ \frac{\beta_n}{\beta_n - \alpha_n} \right\}$ is bounded, then $\lim_{n \rightarrow \infty} D_n = f'(0)$

3) If f' is continuous in $(-1, 1)$, then $\lim_{n \rightarrow \infty} D_n = f'(0)$

4) Find Counter examples.

Proof. Observe that

$$\begin{aligned} \left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(0) \right| &= \left| \frac{f(\beta_n) - f(0)}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(0)}{\beta_n - \alpha_n} - f'(0) \right| \\ &= \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} \cdot \frac{\beta_n}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(0)}{\alpha_n - 0} \cdot \frac{\alpha_n}{\beta_n - \alpha_n} - f'(0) \right| \\ &\leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(0)}{\beta_n - 0} - f'(0) \right| + \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\alpha_n) - f(0)}{\alpha_n - 0} - f'(0) \right| \quad (*) \end{aligned}$$

Now, if $\left\{ \frac{\beta_n}{\beta_n - \alpha_n} \right\}$ is bounded $(*)$ tends to 0 as $n \rightarrow \infty$. $\left(\left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \right)$.

$$\text{If } \alpha_n < 0 < \beta_n, \quad \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \leq \left| \frac{\beta_n}{\beta_n} \right| = 1 \quad \text{and} \quad \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \leq \left| \frac{\alpha_n}{\alpha_n} \right| = 1$$

Thus 1), 2) are proved.

And 3) can be solved by applying MVT. For any $n \in \mathbb{N}$, there exists $c_n \in (\alpha_n, \beta_n)$ s.t.

$$D_n = \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} = f'(c_n)$$

Now, $c_n \rightarrow 0$ as $n \rightarrow \infty$ and f' is continuous,

$$D_n = f'(c_n) \rightarrow f'(0) \quad \text{as } n \rightarrow \infty.$$

Try for 18.

$$\begin{aligned}
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{(\beta_n - \alpha_n) \cdot f'(\alpha)}{\beta_n} \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) - \left(\frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{\alpha_n \cdot f'(\alpha)}{\beta_n} \right) \right| \\
 &\leq \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) \right| + \left| \frac{\alpha_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\alpha_n) - f(\alpha)}{\alpha_n - \alpha} - f'(\alpha) \right|
 \end{aligned}$$

$$\left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \leq M \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \rightarrow M \cdot |f'(\alpha)|.$$

Given $\epsilon > 0$, there exists $\delta > 0$ s.t. $0 < |x| < \delta \Rightarrow \left| \frac{f(x) - f(\alpha)}{x - \alpha} - f'(\alpha) \right| < \epsilon$.
 , since $f'(\alpha)$ exists. Take $N \in \mathbb{N}$ s.t.

$$n \geq N \Rightarrow -\frac{\delta}{2} < \alpha_n < 0 \text{ and } 0 < \beta_n < \frac{\delta}{2}$$

(Note: 여기서 이상함을 느끼게 되는데, 좌변 형태와 다른 적분은 불확실 어떻게 할까?)

Meanwhile, since $f'(\alpha)$ exists, f is continuous at α . Therefore,

$$\lim_{n \rightarrow \infty} f(\alpha_n) = \lim_{n \rightarrow \infty} f(\beta_n) = f(\alpha)$$

$$\begin{aligned}
 \left| \frac{f(\beta_n) - f(\alpha_n)}{\beta_n - \alpha_n} - f'(\alpha) \right| &= \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n - \alpha_n} - f'(\alpha) \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - \frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{(\beta_n - \alpha_n) \cdot f'(\alpha)}{\beta_n} \right| \\
 &= \left| \frac{\beta_n}{\beta_n - \alpha_n} \right| \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} - f'(\alpha) - \left(\frac{f(\alpha_n) - f(\alpha)}{\beta_n} - \frac{\alpha_n \cdot f'(\alpha)}{\beta_n} \right) \right|
 \end{aligned}$$

$$\left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha_n} \right| = \left| \frac{\beta_n}{\beta_n - \alpha_n} \cdot \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \leq M \cdot \left| \frac{f(\beta_n) - f(\alpha)}{\beta_n - \alpha} \right| \rightarrow M \cdot |f'(\alpha)|.$$

Section 5.22. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable s.t. $f'(t) \neq 1$ for all $t \in \mathbb{R}$.

- 1). f has at most one fixed point.
- 2). $f(t) = t + (1 + e^t)^{-1}$ has no fixed point, although $0 < f'(t) < 1$ for all $t \in \mathbb{R}$.
- 3). If $\sup_{t \in \mathbb{R}} |f'(t)| < 1$, then f has a fixed point.

Proof. Let α, β be two distinct fixed points. Then, by the Mean-value Theorem,

$$\frac{f(\beta) - f(\alpha)}{\beta - \alpha} = 1 = f'(c) \quad \text{for some } c.$$

It is contradiction. 2) is trivial. $f(t) - t = \frac{1}{1 + e^t} \neq 0$. But

$$f'(t) = 1 - (1 + e^t)^{-2} \cdot e^t = 1 - \frac{e^t}{(1 + e^t)^2}$$

is contained in $(0, 1)$. Let's show 3), The Banach-fixed point Theorem.

Solve in next page

Problem. Let $f: I \rightarrow I$ be a differentiable with $\sup |f'(x)| < 1$.

Then f has a fixed point.

Proof. Construct a sequence. Let $x_1 = \alpha \in I$, $x_{n+1} = f(x_n)$ $n \geq 1$.

Take $\beta = \sup |f'(x)| < 1$. Given $n \in \mathbb{N}$, if $x_n = x_{n-1}$ then x_n is the fixed point. If $x_n \neq x_{n-1}$,

$$|x_{n+1} - x_n| = |f(x_n) - f(x_{n-1})| = |x_n - x_{n-1}| \cdot |f'(c)| \leq \beta \cdot |x_n - x_{n-1}| \quad \text{by the MVT.}$$

$$\begin{aligned} \text{Now, } |x_{n+1} - x_n| &= |f(x_n) - f(x_{n-1})| \leq \beta |x_n - x_{n-1}| \\ &= \beta \cdot |f(x_{n-1}) - f(x_{n-2})| \leq \beta^2 |x_{n-1} - x_{n-2}| \\ &\vdots \\ &\leq \beta^{n-1} |x_2 - x_1|. \end{aligned}$$

Given $\epsilon > 0$, put $N \in \mathbb{N}$ such that $\beta^{N-1} |x_2 - x_1| < \epsilon \cdot (1 - \beta)$

Then for any $m \geq n \geq N$,

$$\begin{aligned} |x_m - x_n| &\leq |x_m - x_{m-1}| + |x_{m-1} - x_{m-2}| + \dots + |x_{n+1} - x_n| \\ &\leq (\beta^{m-2} + \beta^{m-1} + \dots + \beta^{n-1}) \cdot |x_2 - x_1| \\ &= \beta^{n-1} (1 + \beta + \dots + \beta^{m-n-1}) \cdot |x_2 - x_1| \\ &< \beta^{n-1} \left(\sum_{k=0}^{\infty} \beta^k \right) \cdot |x_2 - x_1| = \beta^{n-1} \cdot \frac{1}{1-\beta} \cdot |x_2 - x_1| < \epsilon. \end{aligned}$$

Therefore $\{x_n\}$ is Cauchy sequence in $\mathbb{R}$, the complete metric space, thus

$$x^* = \lim_{n \rightarrow \infty} x_n$$

exists. Finally, the differentiable map continuous,

$$f(x^*) = f\left(\lim_{n \rightarrow \infty} x_n\right) = \lim_{n \rightarrow \infty} f(x_n) = \lim_{n \rightarrow \infty} x_{n+1} = x^*.$$

□

Note: 수렴사상 $\rightarrow$ MVT $\rightarrow$ difference of Adjacent 가 작아짐으로 같음 $\rightarrow$ 근사하면 $\rightarrow$ 연속함수 $\rightarrow$ 연속수렴보존

If a continuous map $f: \mathbb{R} \rightarrow \mathbb{R}$ satisfies: f has a local minimum at for any $x \in \mathbb{R}$, then f is constant.

Proof. Let $a \in \mathbb{R}$ be given. Consider a set

$$A \stackrel{\text{def}}{=} \{x \in \mathbb{R} \mid f(x) \leq f(a)\} = f^{-1}([f(a), \infty))$$

Since f is continuous, A is closed set. Meanwhile, for any $b \in A$, b is a local minimum:

there exists an open U containing b s.t. $\forall x \in U$, $f(b) \leq f(x)$

But $f(a) \leq f(b)$, thus $U \subseteq A$. Now, A is non-empty clopen, thus $A = \mathbb{R}$.

Finally, for any $a, b \in \mathbb{R}$, $f(a) \leq f(b)$ and $f(a) \geq f(b)$

Section 8.23.

Define $f(x) = \frac{x^3+1}{3}$. It has three fixed points α, β, γ where
 $-2 < \alpha < -1, 0 < \beta < 1, 1 < \gamma < 2$

For arbitrary $x_1 \in \mathbb{R}$, define $\{x_n\}$ as $x_{n+1} = f(x_n), n \geq 1$.

1) If $x_1 < \alpha$, then $x_n \rightarrow -\infty$ as $n \rightarrow \infty$

2) If $\alpha < x_1 < \gamma$, then $x_n \rightarrow \beta$ as $n \rightarrow \infty$

3) If $\gamma < x_1$, then $x_n \rightarrow \infty$ as $n \rightarrow \infty$

Proof. Observe that $f'(x) = x^2$. And, for each $n \geq 1$,

$$x_{n+1} - x_n = f(x_n) - x_n = \frac{x_n^3 + 1}{3} - x_n$$

Suppose that $x_1 < \alpha$. For $n=1$, $x_2 - x_1 = f(x_1) - x_1 < 0$.

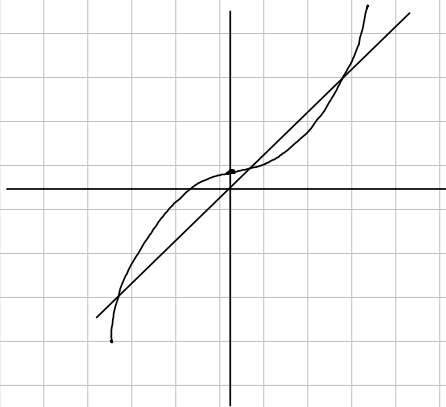
And, if $x_n - x_{n-1} < 0$, then for some t_n between x_n and x_{n-1} ,

$$x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1}) < 0 \quad \text{by } f'(t_n) > 0.$$

Inductively, $\{x_n\}$ is decreasing. Moreover,

$$x_{n+1} - x_n = f(x_n) - f(x_{n-1}) = f'(t_n)(x_n - x_{n-1}) \leq x_n - x_{n-1} \quad \text{by } f'(t_n) > 1.$$

Therefore, $x_n - x_1 = \sum_{k=1}^{n-1} x_{k+1} - x_k \leq (n-1)(x_2 - x_1) \rightarrow -\infty$ as $n \rightarrow \infty$,
thus $x_n \rightarrow -\infty$. Similarly, $x_n \rightarrow \infty$ if $x_1 > \gamma$.



To. 정공헌

정공헌

해당하세요

2026. 05. 22 금
시립대 시의 6년 에세